

Relation Algebra

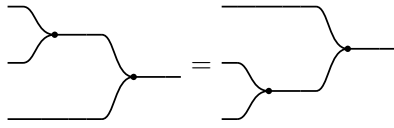
§2. Relations

(2a)

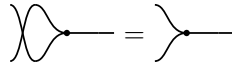
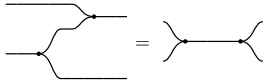
$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ \dashv \circ$, and

$C: (\triangleright : A \otimes A \rightarrow A, \multimap : I \rightarrow A)$ a commutative monoid with $C^\circ \dashv C$. We write C° as $(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow I)$

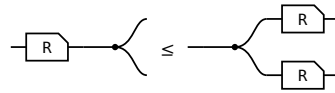
(2b)



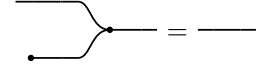
$\triangleright$ associative



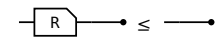
$\triangleright$ commutative



$R \triangleleft \leq \triangleleft (R \otimes R)$



$\multimap$ is its unit



$R \multimap \leq \multimap$

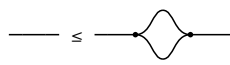
Frobenius, one half — the other is its $\circ$

§2.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$

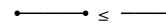
(2.1a)



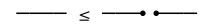
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft \triangleright$



$\multimap \multimap \leq 1$



$1 \leq \multimap \multimap$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.